

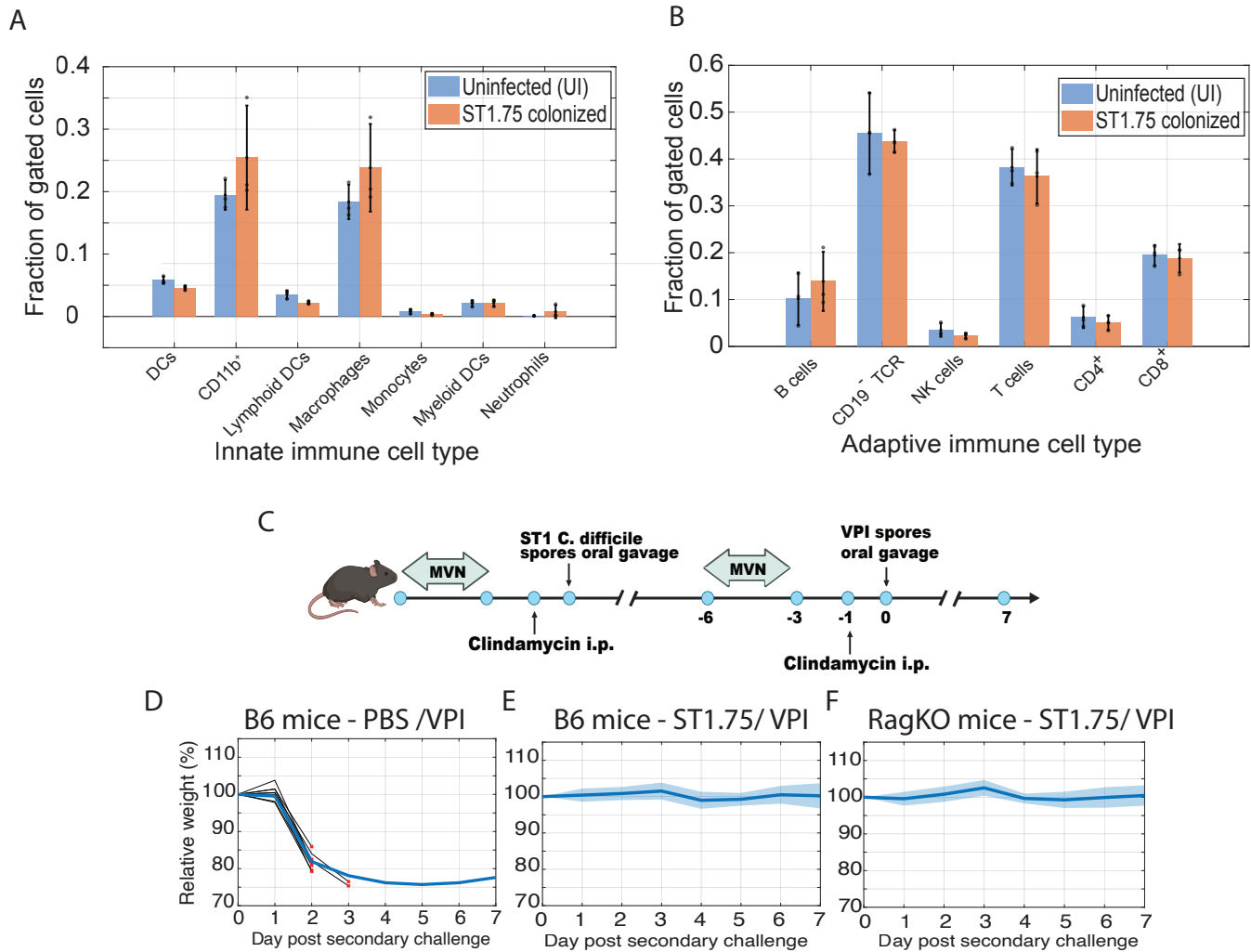


Figure 2: Protection conferred by ST1-75 is independent of adaptive immunity. (A) Fraction of gated cells for innate immune cell populations in the colon and mesenteric lymph nodes of uninfected (UI, blue) and ST1-75-colonized (orange) mice, quantified by multi-parameter flow cytometry. Cell types shown include DCs, CD19-TCR β - cells, NK cells, CD11b⁺ cells, macrophages, monocytes, and neutrophils. No significant differences were observed between groups (Wilcoxon rank-sum, $n = 3$ per group, all $p \geq 0.1$). Error bars indicate SEM. (B) Fraction of gated cells for adaptive immune cell populations in the colon and mesenteric lymph nodes of uninfected (UI, blue) and ST1-75-colonized (orange) mice. Cell types shown include B cells, CD4⁺T cells, CD8⁺ T cells, T cells, $\gamma\delta$ T cells, and DCs. No significant differences were observed between groups (Wilcoxon rank-sum, $n = 3$ per group, all $p \geq 0.1$). Error bars indicate SEM. (C) Schematic of the three-arm experiment. C57BL/6 mice received either PBS (sham) or ST1-75 spores as the primary infection, and RAG1^{-/-} mice received ST1-75 as the primary infection. All groups were subsequently rechallenged with VPI10463 spores following the same MVN antibiotic and clindamycin regimen. (D-F) Relative weight trajectories following VPI10463 secondary challenge for (D) PBS-pretreated B6 mice (sham), (E) ST1-75-precolonized B6 mice, and (F) ST1-75-precolonized RAG1^{-/-} mice ($n = 10$ per group). Shaded regions indicate 95% confidence intervals.